

**LAND
OF THE
CURIOUS**

Service Design

Annika Wolff, Tanvir Hasan

*Professor of Software Engineering,
Researcher in Software Engineering*





Chapter 0

Organizational aspects and general information

Section: How it work's

Professor Annika Wolff

Main teacher in charge

Tanvir Hasan (2nd teacher in charge)

- Studies: Masters in Software Engineering at LUT University, PhD in Software Engineering at LUT University
- TA in course Models and Modelling
- 2nd teacher in course User Experience Design
- 5+ years of experience working in Horizon Europe projects
- Junior research assistant at LUT (Project: CroBoddit)
- Junior researcher at LUT (Project: Mukkula, ParCos, TANDEM)
- Project researcher at LUT (Project: TANDEM)
- Researcher at LUT (Project: CELINE)
- Project manager (Project: TANDEM, CELINE, AMIGOS, TIPS4PED)

Dissertation: *"Integrating User-Centered Design into Scrum: A Canvas Approach"*

General information / How it works

My Research

User Centered Design (UCD)

Software Development Process (Scrum)

How to integrate UCD design in Scrum

User involvement in Software development

Co-design, Living Lab (HE projects)

Data sense-making



Teaching assistant

Essi Aittamaa (essi.aittamaa@student.lut.fi)

- Doctoral student at LUT University, Department of Software Engineering. Researching how to design and develop intelligent and AI-based **characters** that can represent organizations in **digital worlds**
- Screenwriter

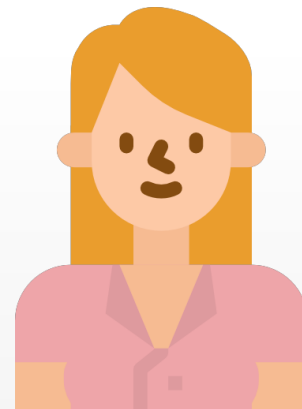
Research articles

- Aittamaa, E., Siemon, D., & S. Singh (2026). Design Elements for Brand Representation in Conversational Agents. HICSS 2026 Proceedings
- Aittamaa, E., & Siemon, D. (2024). Workshops to design conversational agents which embody brand. AMCIS 2024 Proceedings.
- Siemon, D., & Aittamaa, E. (2023). Designing Intelligent Characters Representing Organisations in Digital Worlds. AMCIS 2023 Proceedings.



That's me – Introduce yourself

- Tell us your name and where you are from!
- What do you study?
- Do you work? Tell us more about it!
- Your hobbies and interests
- What ever you want to tell us 😊



Goals and philosophy of my teaching

- No monologue of the teacher
 - Interact, discuss and argue
 - Be able to present your own opinion
 - Tell us about your experience and knowledge
- Dealing with complex problems and issues
- Oral presentation skills and teaching skills
- Reflection, critical thinking and questioning skills

21th century skills

Deeper learning

Happiness

Digital literacy

Digital society

Complex problem solving

Teamwork

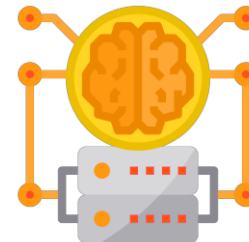
Collaboration

Leadership

Applied skills, people skills, social skills,
life skills

Title

- Relevant bullet points and explanations
- Another important bullet point
- Etc.



*A representative icon to
help the building of
associations*

DEF

Here is a definition of a term relevant to the chapter.

S

Autor et al. (2019)

KM: Key message of that slide



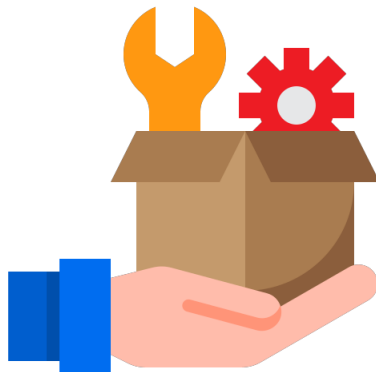
Chapter 0

Organizational aspects and general information

Section: Overview and content

- What I think the course is about (what I expect to learn)
- How I think I can contribute to the course
- How I think I can have fun 😊

Service Design



Service Design

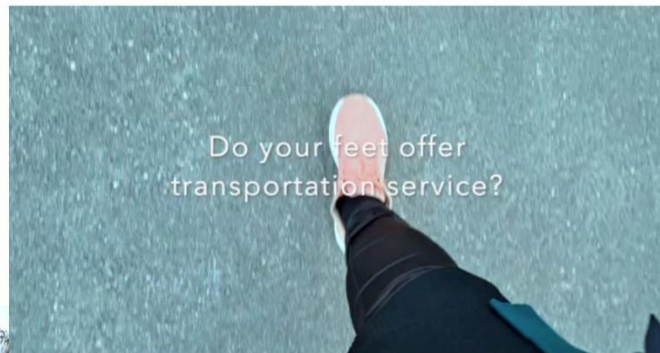
Value

What is a service?
How is it different to a product?
Why are services so important?
What does services have to do with software?

What is design?
Where is design relevant?
How to do good design?
How to do systematic design?

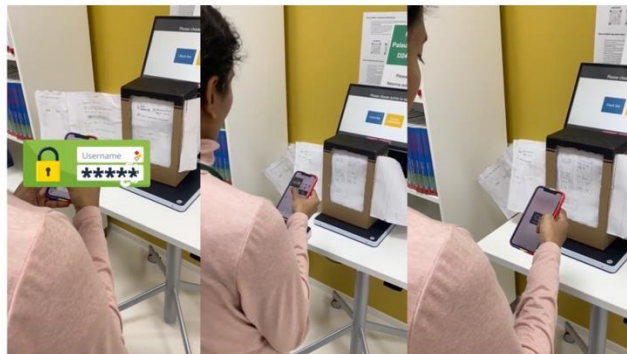
How to design for service?

Service Design isn't sitting in a room

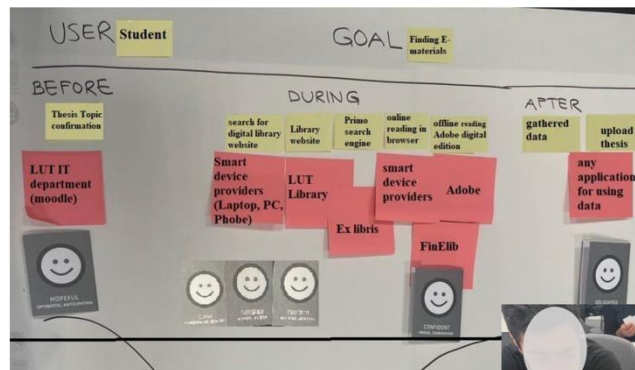


Out of the classroom!

Photo credits: students of Autumn 2023



Service Design is hands on!

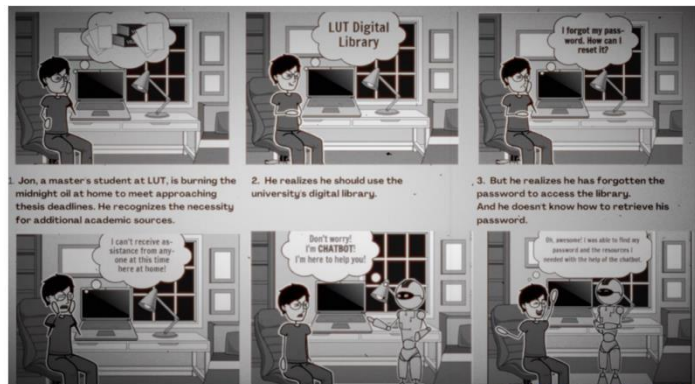


Mapping out what your learn from observations, desk research, interviews!



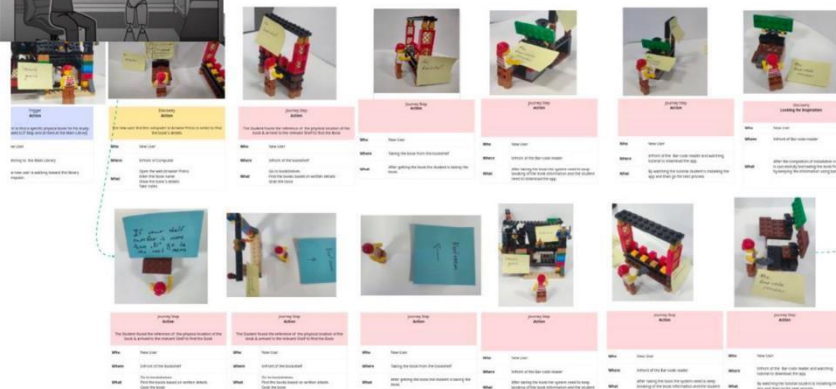
Photo credits: students of Autumn 2023

Service Design is creative!



Creating user stories and scenarios!

Photo credits: students of Autumn 2023



Service Design is experimenting!



Service Design is collaborating!



**A lot of group work during
class hours and outside of
them!**



Photo credits: students of Autumn 2023

Lectures, tutorial and project

1. Understand the difference between a product and a service (do products exist?)
2. Know the characteristics of a service and understand the importance of services
3. Understand what design is and what forms design can have
4. Understand the concept of value and value creation
5. Being able to capture the *proposed* value of users
6. Being able to design services with a value for its users in mind



Service Design

1. Principles and Theories of Services

Service

1. Introduction and Service Definition
2. Service-dominant Logic and Value
3. Digital Services and Digitalisation of Service

2. Principles and Theories of Design

Design

1. Introduction to Design
2. Design Thinking
3. Design Science
4. Participatory Design

3. Principles and Theories of Service Design

Service Design

1. Definition of Service Design
2. Human-centred
3. Collaboration and Co-creation
4. Iterative and Sequential
5. Real and Holistic
6. Service Design Thinking and Processes

4. Tools and Methods of Service Design

Methods/Tools

1. Research
2. Mapping Stakeholders and Ecosystems
3. Journey Maps, Touchpoints and Service Moments
4. Service Blueprint
5. Ideation
6. Prototyping



Grading

Your grade for this course will consist of the **sum of grades** received for the following activities. Please note that most of the grades come from **group work**, so pay attention to the group you will form at the beginning of the course, as you will stay in the same group the whole semester.

1. 12 x TUTORIAL EXERCISES (15% of overall grade) for submitting on time and having followed
2. instructions. These will take place almost every week.
3. 2 x WRITTEN GROUP ASSIGNMENTS (GR1 + GR2) (15% + 25% of overall grade). **Submit one per group of each report.**
4. 2 x ORAL GROUP PRESENTATIONS (GP1 + GP2) (10% + 10% of overall grade). **Oral presentations are recorded and uploaded**
5. 2 x INDIVIDUAL REFLECTION written report (IR1 + IR2) (10% + 15% of overall grade).
6. ADDITIONAL 5% available (if you submit all 12 of the tutorial exercises on Moodle)

Please note that the tutorial exercises will take place throughout the semester; GR1, GP1, and IR1 have a deadline after the teaching break and GR2, GP2, and IR2 at the end of the semester.

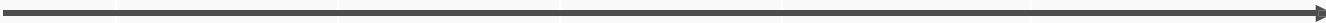
NO EXTENSIONS FOR COURSEWORK WITHOUT PRIOR AGREEMENT: Agreement only given in exceptional circumstances.



Timetable part 1 – Lectures and Tutorial

Week	Week 1 31.8.+2.9.	Week 2 7.9.+9.9.	Week 3 14.9.+16.9.	Week 4 21.9.+23.9.	Week 5 28.9.+30.9.	Week 6 5.10.+7.10.	Week 7 12.10.+14.10.
Lecture topic	1. Principles and theories of services			2. Principles and theories of design		3. Principles and theories of service design	
Sub-topic	Introduction and service definition	Service-dominant logic and value	Digital services and digitalisation of service	Introduction to design	Design thinking, design science, participatory design	Service design	Service design thinking and processes
Exercise	Tutorial 1	Tutorial 2	Tutorial 3	Tutorial 4	Tutorial 5	Tutorial 6	Tutorial 7
Assignments	Start GP1+GR1 Start <						

Timetable part 1 – Lectures and Tutorial

Week	Week 8 26.10.+28.10.	Week 9 2.11.+4.11.	Week 10 9.11.+11.11.	Week 11 16.11.+18.11.	Week 12 23.11.+25.11.	Week 13 30.11.+2.12.	Week 14 7.12.+9.12.
Lecture topic	1. Tools and methods of service design and project						
Sub-topic	Research	Mapping stakeholders and ecosystems	Journey maps, touchpoints and service moments	The service blueprint	Ideation	Prototyping	Prototyping
Exercise	Tutorial 8	Tutorial 9	Tutorial 10	Tutorial 11	Tutorial 12	No tutorial	No tutorial
Assignments	Start  GP2+GR2+IR2						

AI Statement

You are required to include an "artificial intelligence statement" in the **EVERY DOCUMENT YOU HAND IN** (this is not counted towards the length limit).

Otherwise you get 0 points on this document

AI Statement

- No content generated by AI tools has been presented as my own work.
- I acknowledge the use of [AI system and link] to generate materials for background research in drafting this assessment.
- I acknowledge the use of [AI system and link] to generate materials that were included within my final assessment in modified form. The following were initially generated by AI: <Provide a bullet point list of materials and code and explain the modifications you have made>.

You must include this statement, even if you did not use AI.

Further information about the University guidelines are available here:

<https://elut.lut.fi/en/completing-studies/rules-and-regulations/ai-based-tools-policies>



Chapter 0

Organizational aspects and general information

Section: Thought starter

Thought starter

“Companies don’t deliver value, consumers create value!”

Christian Grönroos

What does this mean for a car?

What does this mean for software?

5 Minutes individual thinking
Discussion

Break





Chapter 1

Principles and theories of services

Section: Characteristics of Services

WHAT COMES TO YOUR MIND... ... when you hear the words "service"?
Write the name of an example of a service that you use:

Join at menti.com | use code 3109 4640

WHAT COMES TO YOUR MIND... ... when you hear the words "service"?

All responses to your question will be shown here

Each response can be up to 200 characters long

Turn on voting to let participants vote for their favorites



DS



Menti

Service Design



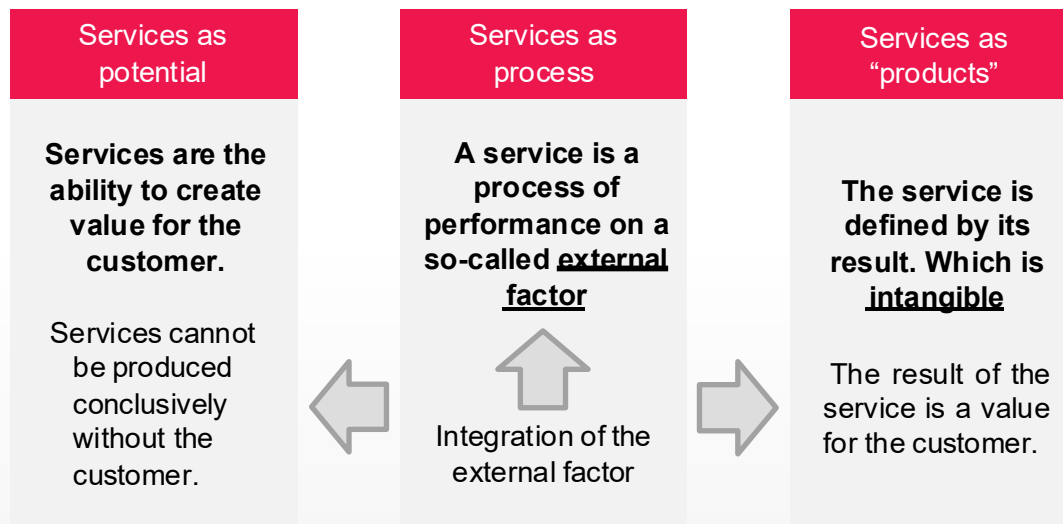
Choose a slide to present

Expectations

WHAT COMES TO YOUR MIND... ... when you hear the words "service"?

Definition of a service

Services are something that may be bought and sold, but which cannot be dropped on your foot.



e.g. accounting, banking, cleaning, consultancy, education, insurance, medical, or transportation...

Diversity of services

- **Hands-on services** (janitors, teachers, doctors...)
- **Technology-mediated services** (self-service check-outs at stores, online travel reservations, package delivery...)
- **Simple and complex services** (e.g. massage therapist services, accounting firm services)
- **Local or geographically** distributed (car rental services)
- Different skill levels required from the providers and recipients of services

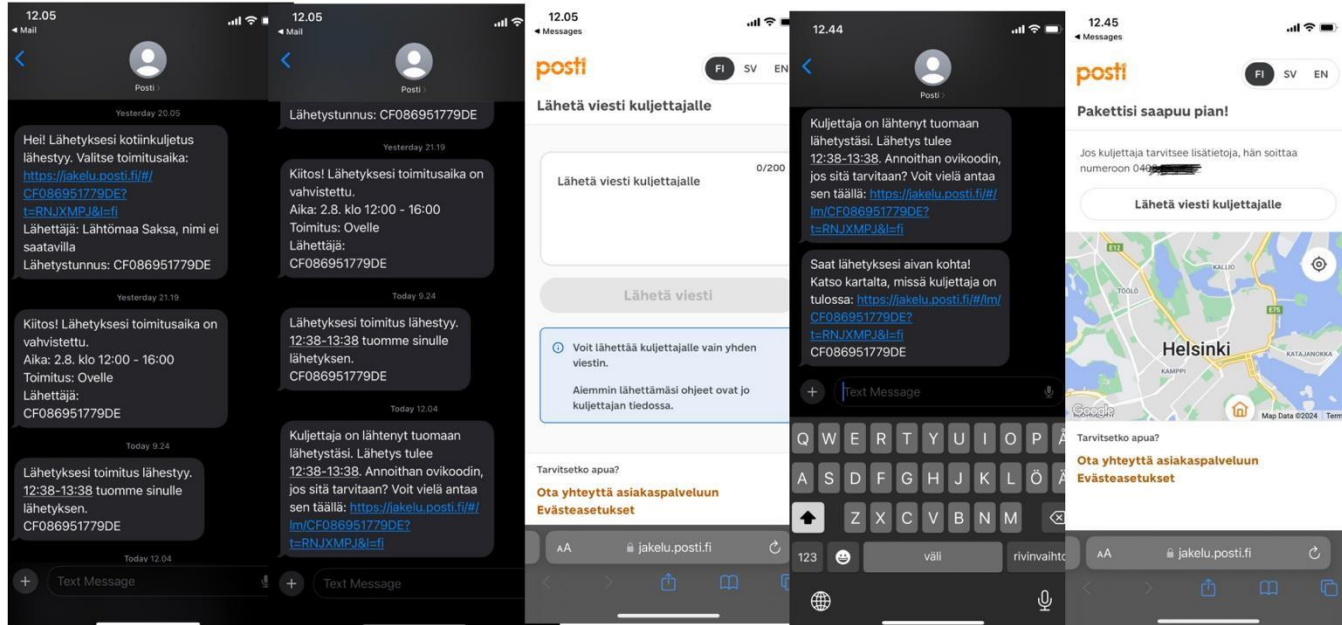


(Blomberg and Darrah, 2015)

Example: posti

Example by Joanna Saad-Sulonen

Choose delivery timeslot > narrower timeslot > delivery person on its way + can be contacted > follow delivery van > package delivered ☺



Service offering

- Set of services, features, benefits provided by a business or organisation for its customers to meet their needs or solve their problems

Provide CARE, ACCESS, or
RESPONSE to people or
things

3 key primary spheres of service offerings



Products vs services

Aspect	Product	Service
Nature	Tangible, physical	Intangible, experiential
Ownership	Customer owns after purchase	No ownership, access or use only
Storage	Can be stored and inventoried	Cannot be stored, produced and consumed simultaneously
Standardization	Standardized, reproducible	Variable, depends on provider and context
Customer role	Passive receiver	Active participant in co-creation
Value creation	Embedded in the product itself	Created during delivery and interaction

(Kotler & Keller 2016)

Services and products

Product ↔ Service

» Downe: “Services can be thought of in conjunction with products: they exist around the product”



So, the service is not the sandwich per se but the process of being welcomed when one is hungry, being given options to choose from, ordering the sandwich of your choice, waiting for it, receiving a good and clean sandwich, taking the first bite ...

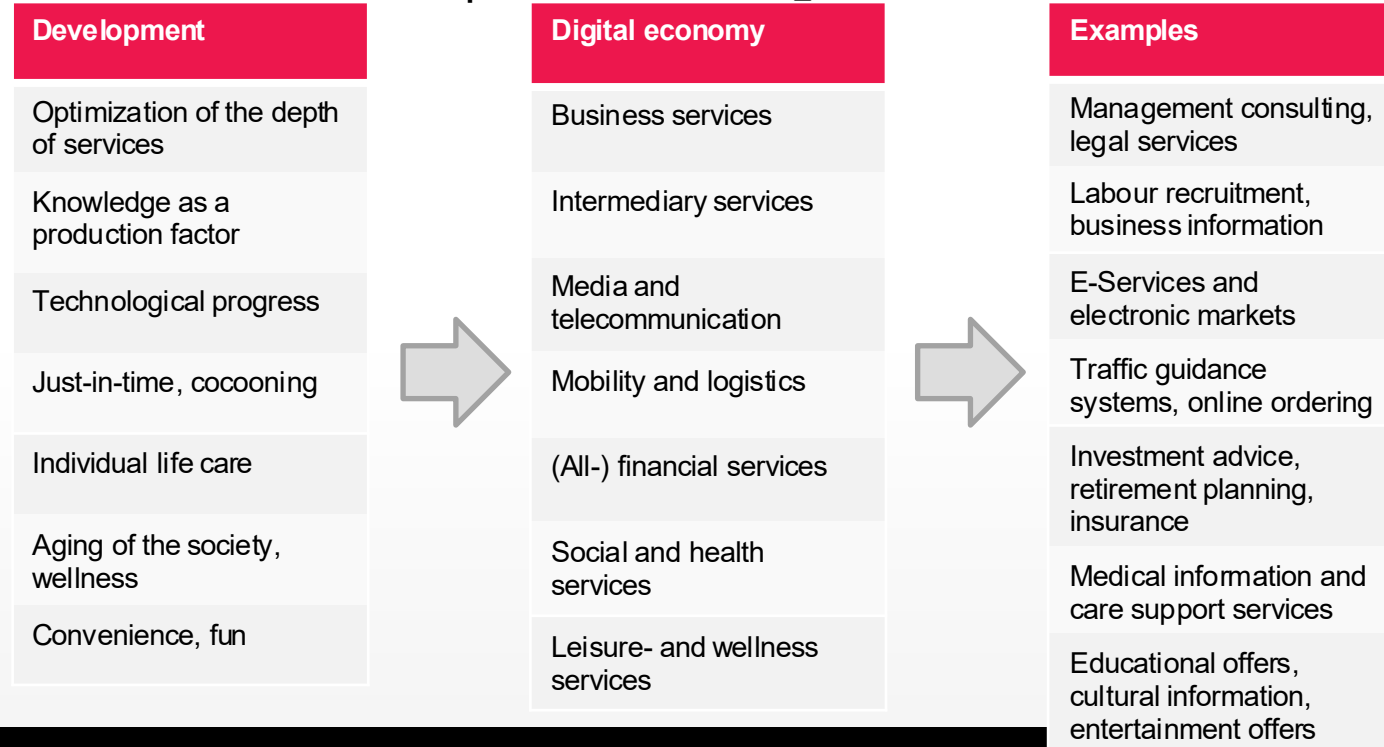
Product/service hybrids



- Miettinen: Companies developing products are also increasingly developing hybrid products (increase profitability, growth, customer retention) → service is designed as an inseparable part of the product
- Sangiorgi & Prendiville: Products and services increasingly hard to separate in contemporary organisations
- Blomberg and Darrah: Information technology allows hyper-customisation of products
- Blomberg and Darrah: Products that we once standalone are now services

Societal development: Service economy

Social developments lead to growth areas in services



Servicification: From goods to value

Servicification: The offer portfolio shifts away from just tangible goods and towards a combination of tangible goods and services.



Food

Trade

Information

Buying: Own food



Meals

Delivery

Digital connection

Easily eat at home



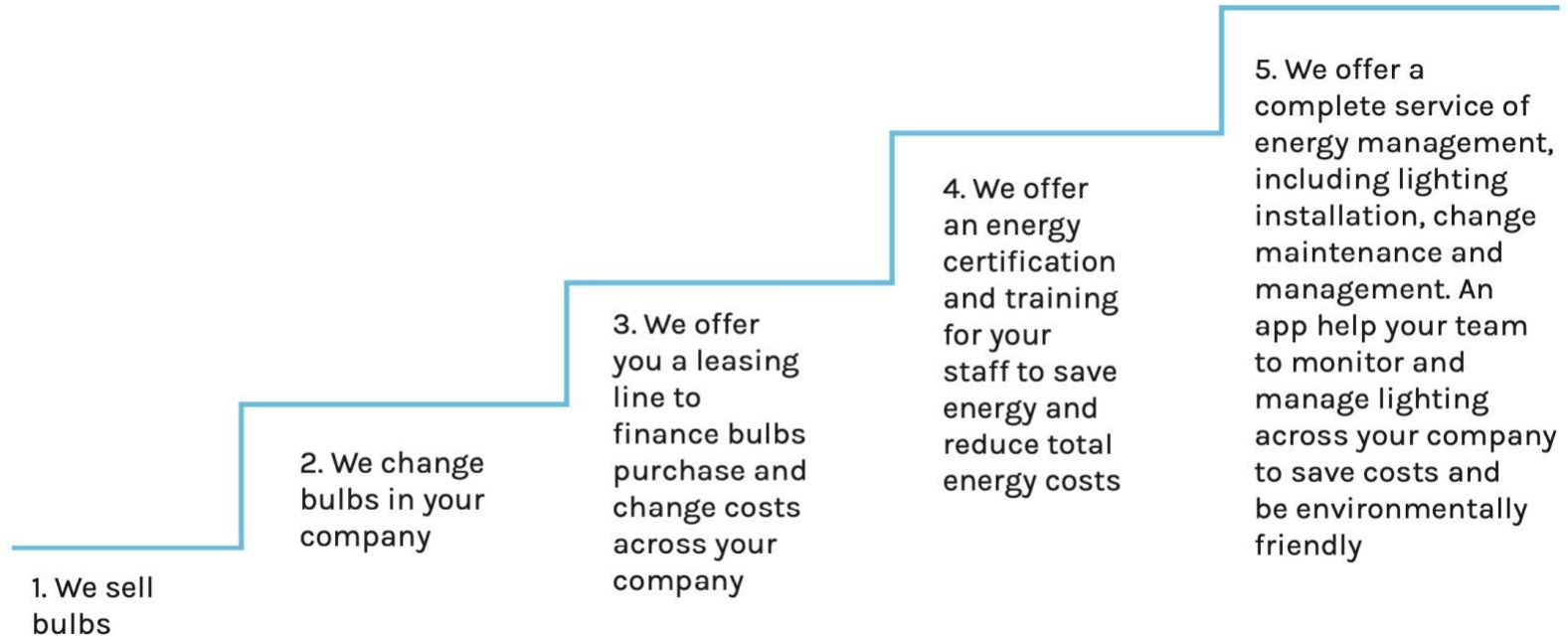
Ingredients/Recipe

Composition/Delivery

Digital selection

Cooking fast and refined

Servicification: Bulb example



Three (four, five) sectors economy

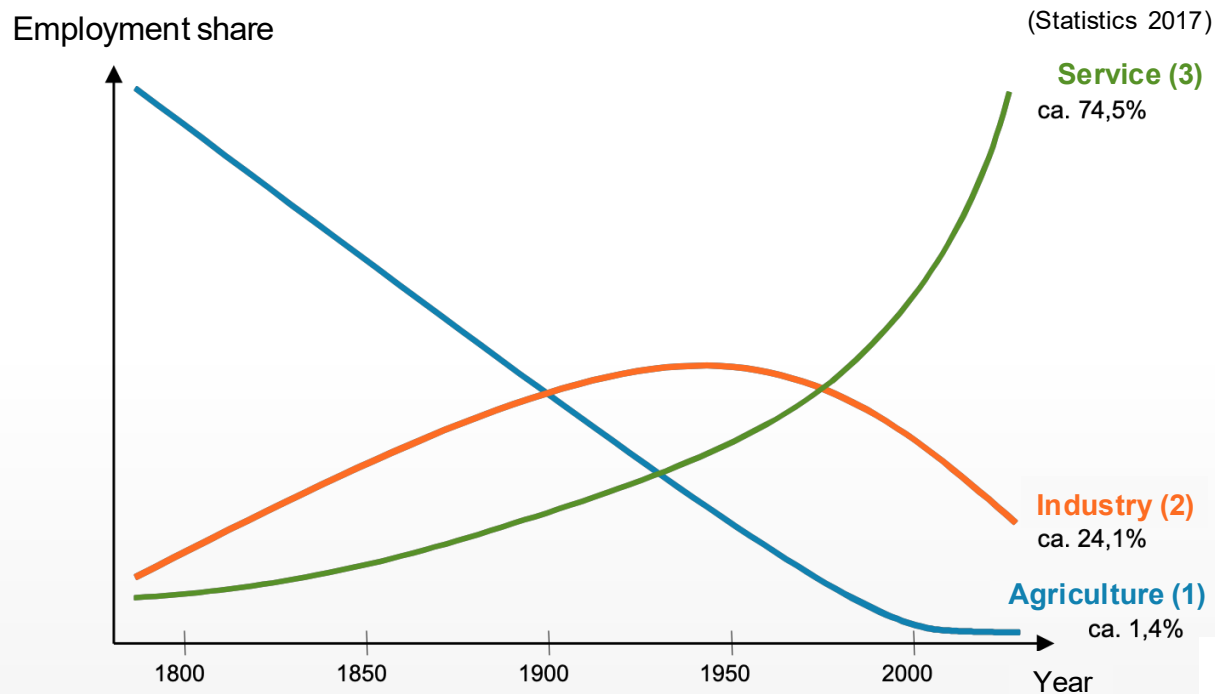
Sectors of the economy	
Primary (raw materials)	Extraction of raw materials Farming/fishing Agriculture
Secondary (finished goods)	Manufacturing Utilities, electricity, gas Construction
Tertiary (service sector)	Retail Financial services Communication Hospitality and leisure Real estate Information technology



Quaternary sector

Quinary sector

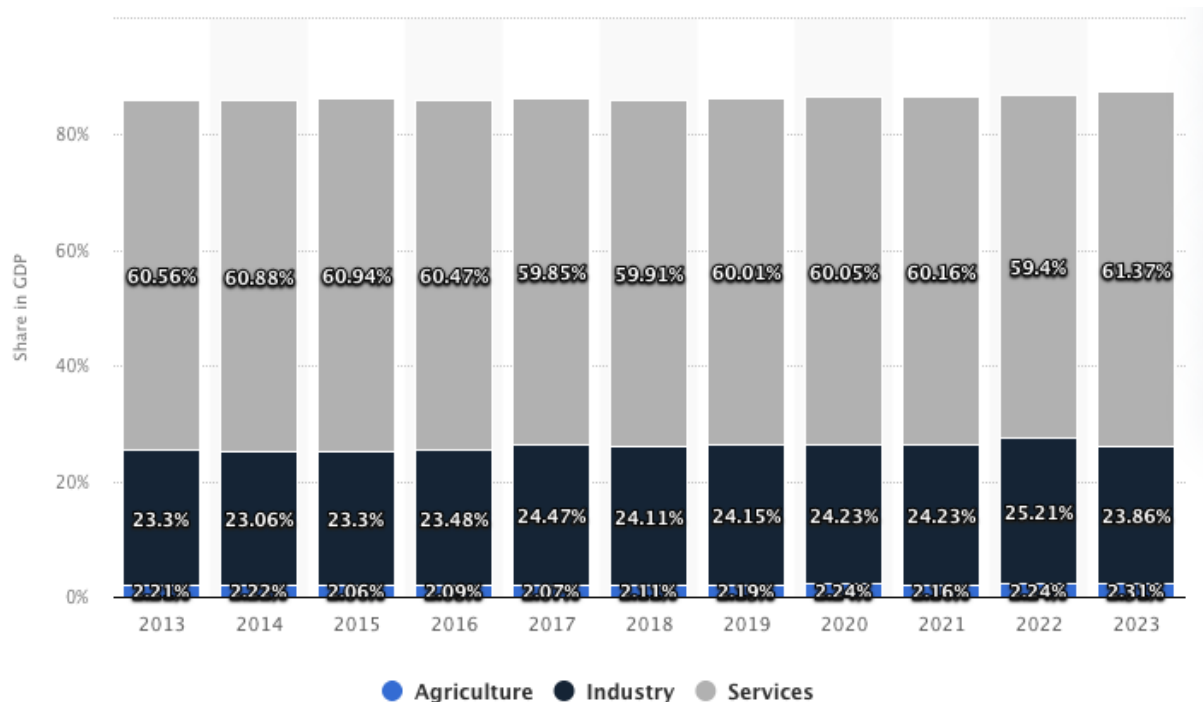
Relevance of services (three sectors)



KM: Services are getting more and more relevant

**Just a visualization*

Finland: Share of economic sectors

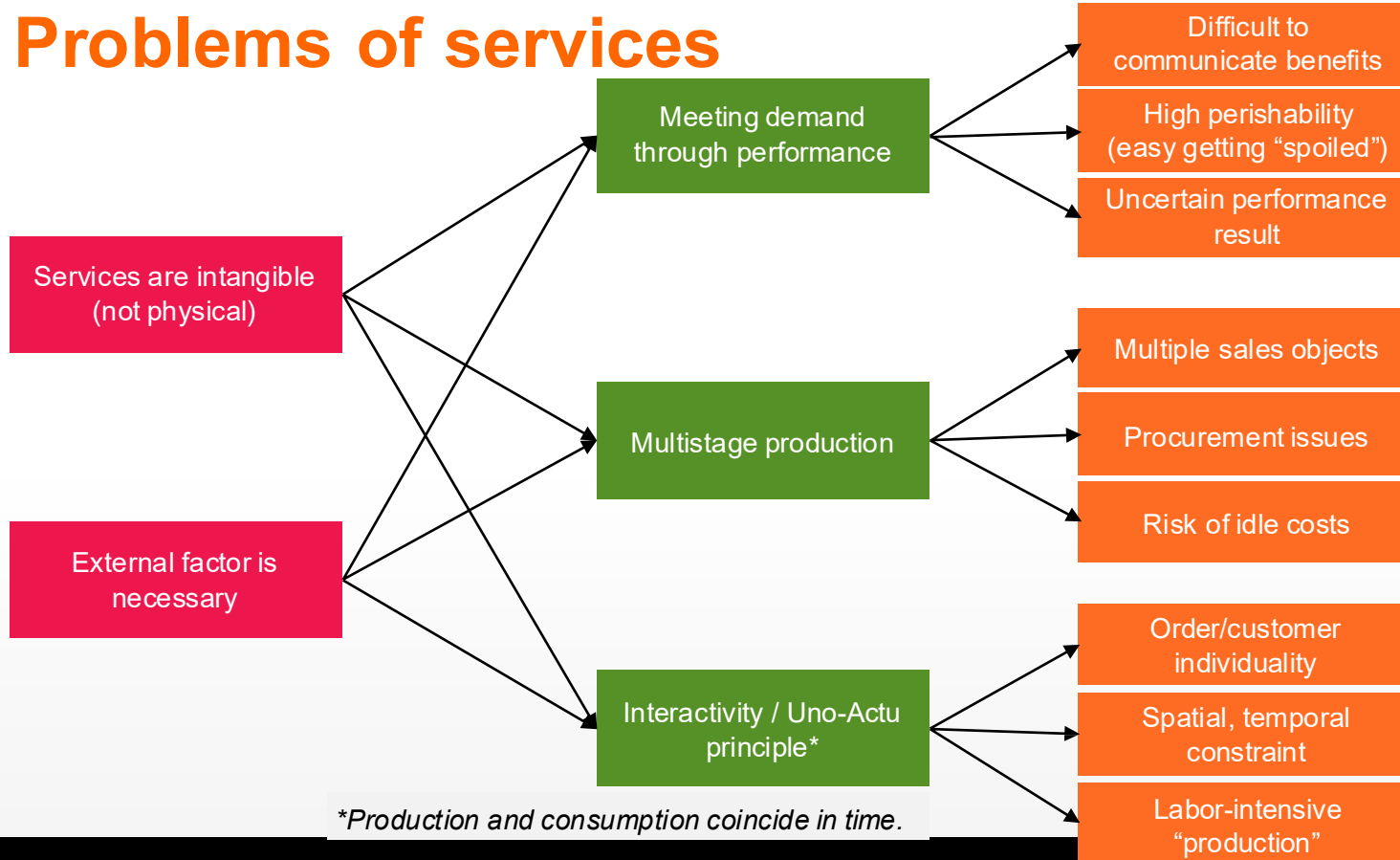


Source
World Bank
© Statista 2023

GDP: Gross Domestic Product

KM: Services are super important

Problems of services



The service process (traditional view)

“Service value chain”

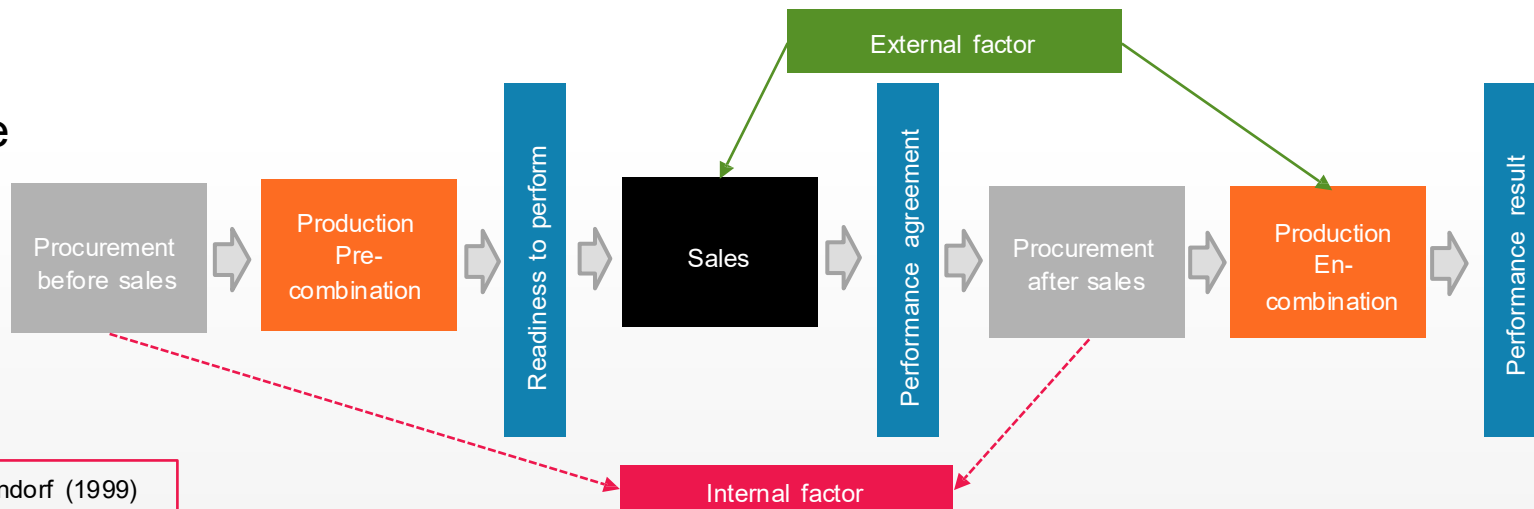
Tangible asset

Procurement

Production

Sales

Service



S

Bodendorf (1999)

Internal factor